Data Analysis and Python Code

26/03/2025

HIV_Epidemiology.... .csv
19.2 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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To analyze the dataset and provide a data-driven response, I will first load the dataset and examine its structure, including column names, data types, and any missing values. Then, I will address a specific analytical question using Python code.

Step 1: Load and Explore the Dataset

First, let's load the dataset and inspect its key characteristics.

python
```python
import pandas as pd

# Load the dataset
file_path = 'HIV_Epidemiology_Children_Adolescents_2024_out.csv'
data = pd.read_csv(file_path)

# Display basic information about the dataset
print("Dataset Overview:")
print(data.info())

# Display the first few rows of the dataset
print("\nFirst 5 Rows:")
print(data.head())

# Check for missing values
print("\nMissing Values per Column:")
print(data.isnull().sum())
```

Step 2: Address a Specific Analytical Question

Since no specific question has been provided yet, I will propose a potential question based on the dataset's likely content (e.g., HIV epidemiology in children and adolescents).

Proposed Question:
"What is the total number of HIV cases reported among children and adolescents in the year 2023?"

Here's how we can answer this question:

python
```python
# Ensure the dataset contains relevant columns such as 'Year' and 'Cases'
if 'Year' in data.columns and 'Cases' in data.columns:
    # Filter the data for the year 2023
    data_2023 = data[data['Year'] == 2023]

    # Calculate the total number of cases
    total_cases_2023 = data_2023['Cases'].sum()

    print(f"\nTotal HIV Cases Reported in 2023: {total_cases_2023}")
else:
    print("\nThe dataset does not contain the required columns ('Year' or 'Cases') to ar
```

Deliverables

1. **Answer to the Question:** The total number of HIV cases reported among children and adolescents in 2023.
2. **Python Code:** Provided above, which includes loading the dataset, exploring its structure, and calculating the total cases for 2023.

If you have a different or more specific analytical question in mind, please provide it, and I will tailor the analysis accordingly.

Is there an association between mother-to-child HIV transmission rates and changes in HIV-associated mortality at the national level from 2005 to 2023?

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To address the question of whether there is an association between mother-to-child HIV transmission rates and changes in HIV-associated mortality at the national level from 2005 to 2023, we will follow these steps:

Step 1: Understand the Dataset

The dataset must contain columns that represent:

1. **Mother-to-child HIV transmission rates** (e.g., percentage or counts).
2. **HIV-associated mortality rates** (e.g., deaths per 100,000 population or total counts).
3. **Year** (to filter data from 2005 to 2023).
4. **Country or National Identifier** (to analyze trends at the national level).

If these columns are not present, the analysis cannot proceed as intended.

Step 2: Analyze the Association

We will calculate the correlation between mother-to-child transmission rates and HIV-associated mortality rates for each country over time. A correlation coefficient (e.g., Pearson's r) will indicate the strength and direction of the association.

Step 3: Python Code

Below is the Python code to perform this analysis:

```python
import pandas as pd
import numpy as np

# Load the dataset
file_path = 'HIV_Epidemiology_Children_Adolescents_2024_out.csv'
data = pd.read_csv(file_path)

# Check if required columns exist
required_columns = ['Country', 'Year', 'Mother_to_Child_Transmission_Rate', 'HIV_Mortali
if not all(col in data.columns for col in required_columns):
    print("The dataset does not contain the required columns to answer the question.")
else:
    # Filter data for the years 2005 to 2023
    filtered_data = data[(data['Year'] >= 2005) & (data['Year'] <= 2023)]

    # Group by country and calculate correlation for each country
    correlations = []
    for country, group in filtered_data.groupby('Country'):
        # Drop rows with missing values for the two variables
        group = group[['Mother_to_Child_Transmission_Rate', 'HIV_Mortality_Rate']].dropn

        # Calculate correlation if there are enough data points
        if len(group) > 1:  # At least 2 data points are needed for correlation
            correlation = group['Mother_to_Child_Transmission_Rate'].corr(group['HIV_Mor
            correlations.append({'Country': country, 'Correlation': correlation})

    # Convert results to a DataFrame
    correlation_df = pd.DataFrame(correlations)

    # Display summary statistics of correlations
    print("\nSummary of Correlation Coefficients:")
    print(correlation_df['Correlation'].describe())

    # Identify countries with strong positive or negative correlations
    strong_positive = correlation_df[correlation_df['Correlation'] > 0.7]
    strong_negative = correlation_df[correlation_df['Correlation'] < -0.7]

    print("\nCountries with Strong Positive Correlation (> 0.7):")
    print(strong_positive)

    print("\nCountries with Strong Negative Correlation (< -0.7):")
    print(strong_negative)
```

Step 4: Interpretation of Results

1. **Correlation Coefficient (`r`)** :
   - Values close to `1` indicate a strong positive association.
   - Values close to `-1` indicate a strong negative association.
   - Values near `0` indicate no significant association.
2. **Summary Statistics** :
   - The mean and standard deviation of the correlation coefficients across all countries provide an overall sense of the relationship.
3. **Strong Associations** :
   - Countries with correlations above `0.7` or below `-0.7` are flagged for further investigation.

Deliverables

1. **Answer to the Question** :
   Based on the dataset, we will determine whether there is a statistically significant association between mother-to-child HIV transmission rates and HIV-associated mortality rates at the national level. For example:
   - If the mean correlation coefficient is close to `0` , there is no general association.
   - If many countries show strong positive or negative correlations, it suggests a potential relationship.
2. **Python Code** : Provided above, which calculates correlations and identifies countries with strong associations.

Next Steps

- If the dataset lacks the necessary columns, please provide clarification or additional data.
- If you have specific countries or regions of interest, let me know, and I can refine the analysis accordingly.